An Empirical Study on Options Trading Strategy Using 'Commodity Channel Index' For NSE's Nifty Options in India

Pinkal Shah[a]
[a]Professor, Department of Management, Sumandeep Vidyapeeth, Vadodara, Gujarat

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ABSTRACT
Though volume of option trading, specifically in NSE'S NIFTY Index contract, rise significantly over a period of time, it is considered as comparatively complicated due to complexities involve in option. pricing. Volatility in prices play very critical role in option pricing. With help of momentum indicators of technical analysis, one can identify point of expected high amount of volatility in prices. This lead to very high amount movement in optin prices. Commodity Channel Index (CCI) is one of the tools of technical analysis which helps in assessing momentum and give hint about future volatility in prices. The present study aims to trace significant (atleast double) rise in option price with minimum risk using CCI. For this purpose, historical data of NSE's NIFTY index was tested on stipulated rules of CCI indicating strong momentum and accordingly creating long position in corresponding in The Money option contracts. The risk and return of using this trading mechanism in NIFTY option was calculated using Return, Maximum Loss Zone, Average Gain to Average Loss Ratio, Strike Rate (Success rate) and its significance. The results are found positive about CCI based trading in option as per stipulated in rules for aggressive traders with time frame of two weeks.

Introduction

The fundamental theory of macroeconomics subscribes the fact that investment is considered as lifeblood for economic system while financial market is considered as blood-vessel. Any organized economy provides different alternatives for prospective investors to invest their saving in monetary form. These alternatives of investment are classified into following two categories viz. Risk free investment alternatives & Risky investment alternatives.

Investment & Speculation in equity shares of different company through stock exchanges, popularly known as investment and trading in stock market is a prominent and widely acceptable alternative in the category of risky investment avenues. Significant proportion of national savings is invested in stock market at global level. In case of India, this proportion is much higher than other developed countries. Indian stock market has witnessed marvellous growth and development of derivative market in last decade in line with increase in market depth and volume in equity market. In Indian derivative market, derivative instruments are available in forms of future and options contracts for different indices and stocks and are available to trade mainly on platform of NSE and BSE. However, it is observed that

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trading in NSE NIFTY's derivative contracts, both future and options are significantly higher in terms of volume and participants compared with other instruments.

Derivative Market & Option Trading in India

In derivative segment, option trading in contracts is considered as comparatively complicated and complex task for any traders. This is due to several dynamic factors affecting option price with different intensity of the option contracts. Moreover, amid volatility or price fluctuation in option contracts is frequently observed. Therefore, it will be difficult for any option trader to establish specific trend unlike in case of stock price in cash segment or future contracts. However, option trading is becoming popular day by day as its awareness and education increases. Further, traders prefer to create long (buy) position in option (either call or put) as it gives leverage position with very small amount of investment. It also actually enables traders to earn large amount of profit with known (limited) loss and with definite exit time.

However, it should be noted that speculation in derivative segment always remains highly risky option in stock market, specifically in option contracts where volatility is significantly higher. Therefore, traders have to follow the scientific approach of security analysis which is part of investment management process. Therefore, prediction of prices for future time frame with expected risk and return is the key to be successful trader in trading with future in stock market. However, predictions about prices in stock market are always challenging as stock prices are fluctuating every second and there are many factors which influence these fluctuations. Therefore, share price prediction always remains as an area of interest amongst investors and researchers. There are different approaches of security analysis for prediction. In real life situation, fundamental analysis and technical analysis are two popular approaches to prediction of share prices. The fundamental analysis mainly answers as to which stocks should be bought or sold for long period of time. It is based on fundamental factors related with economy, industry and company affecting share price movement. Technical analysis mainly deals with the price at which one should enter into the trade and exit the trade with stop loss based on statistical analysis of historical data of share price and its charting pattern.

Technical Analysis & Stochastic Momentum Indicator (CCI)

Technical analysis, though highly criticized by academic approach based on ground of efficient market hypothesis, it remains extremely popular techniques amongst traders and research houses for predicting future prices, specifically for derivative market for last so many years. There are hundreds of indicators in technical analysis which can be used as per their risk and return expectations, their time frame and the investment amount and other trading traits of the investors. However, in this research CCI on NIFTY index is taken into consideration for developing optimum trading system for NSE Nifty options in India. CCI is a lower technical indicator developed by Donald Lambert in 1980 mainly for identifying cycling trend in commodities. However, due to its success in predicting trend, it not only used in commodity but also in equity and forex market widely. It is mainly indicating

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momentum in stock price with high volatility and its trend. Many times, it also gives an advance indication of trend reversal.

The calculation of CCI is as follow:

CCI = (TP – N Period SMA of TP) – (0.015 x MD)

Where,

TP = (High + Low + Close Prices)/3

N = Look Back Period

SMA = Simple Moving Average

MD = Mean Deviation of TP

It should be noted that CCI is typically calculated over 20 days Look Back Period. The value more than 100 signifies overbought situation and less than -100 signifies oversold situation in the security under study. Overbought refers to the price trading near the top of the 20-day range, while oversold refers to the bottom of the range. Overbought and oversold periods can be lengthy, should the price continue to rise or fall.

There are different ways of taking call for buying or selling a security based on this indicator. As per common practice, when CCI goes above -100, a purchase signal is generated. Similarly, when CCI falls below 100, a sell signal is generated. The following figure 1 shows way interpreting CCI in technical chart of NIFTY.

[Figure – I: Interpretation of CCI in Technical Chart]

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About Trading System in Options using Technical Analysis

In a very simple term the trading system is a 'set of pre-defined' rules mainly based on indicator(s) of technical analysis on the basis of which trading in either call or put option is carried out. The term 'predefined' suggests that trading will be planned for well in advance and hence will not be arbitrary or subjective. Similarly, the term 'rules' suggest that the work to be done will be done keeping strictly in mind the parameters laid d own as rules. Thus, it is a set of instructions which advise entry and exit position in option contracts positions based on the results of selected indicator(s) technical analysis.

A trading system allows excluding randomness in the trading process. Strict adherence to the system permits to rule out the emotional factor in the trade. For this reason, one must follow all recommendations of the system strictly even if for all that a potentially profitable position will not be opened. Apart from it, any trading system for option contracts includes trading in both the trend i.e. upward and downward.

Review of Literature

Though technical analysis is a very popular tool for taking trading decisions in equity market, not frequently used in creating long option positions. Further, there are very less amount of literatures found which presents empirical evidences of use of technical analysis for trading in option contracts.

Moving Average is found very popular indicator in technical analysis for developing trading strategy in option. Baja V and Aghav J (2016) also investigates on developing trading system in options of India's NSE NIFTY by using technical indicators viz. Moving Average and Average True Range jointly and individually for the period of 2012-15. They found strike rate of 40% and AGAL with around 4.5 times. However, they have not calculated statistical significance about their system as well as consistency in their return. N M Rao (2013) for NIFTY and selected stock in India but he used Exponential Moving Average as a primary technical indicator for testing bull and bear spread strategies in in India with historical data of 3 years. Further, Chao-Hui Yeh (2012) investigates cross-sectional profitability of technical analysis with special reference to use of moving average in option market of Taiwan. His study shows almost 90% success rate in option positions. In the same line Paul Brittain and Carley Gamer (2006) came out with 8 simple option strategies in which they suggested the trading methods which are based on spreads. But they used trend line method of technical analysis identify support and resistance level for framing option strategies. Charlebois and Sapp (2007) also test Moving Average strategies on daily data on the currency pair of Dollar-Deutsche Mark to take position in either call or put option.

There are few studies which reported use other technical indicators like Bollinger Band and Velocity Index for trading in option. A Kalkova (2016) propose a trading system to trader in currency option by using Bollinger Band to trade in Binary Option. Young Hoon Ko (2011) presented an empirical study on developing straddles

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in options by use of velocity measurement index – an indicator of technical analysis. His study is proposing return of around 1% per trade.

However, no study was found in which Commodity Channel Index (CCI) is used to evaluate trading in options. But, studies show that CCI is very good indicators in spot and future segment of not only equity but also in commodities and forex market. Lakkol S and Meghana A. (2017) investigate use of technical indicators viz. CCI along with KAMA, Moving Average and Bollinger Band for trading in companies who are part of NSE NIFTY index. They found that strict implementation of trading rule of above technical indicators helps trader in generating over performing return from trading with technical indicators. Mahesh M. and Anju M. (2017) investigates on implementing various technical indicators and its ranking for selected companies in Steel Industries in India. Based on historical data of 1 year they found that CCI is amongst top 3 technical indicators who helps investor to gain return higher than market return. Maitah M. et al. (2016) found that swing trading in CCI generate excellent return which outperform normal return by more than 2 time in agricualtural commodities (mainly COCO and Coffee) in Czech Republic. Revathi Bala and Deepak Sundar (2014) studied use of CCI in trading of MCX Crude Oil future contract and found that CCI facilitates in better manner to the commodity trader in to take position in crude oil future. They also noted that CCI is more accurate in predicting downward fall in prices. In similar line P Natarajan and E Nirupama (2014) also concluded that with help of CCI a trader can get abnormal return which outperform normal return in currency market (MCX-SX USD/INR). Ekarin V (2012) uses CCI along with other technical indicators and found around 70% strike rate in selected securities in stock market of Thailand.

So, from literature it can be concluded that there are very few research papers which actually investigates role of technical analysis, specifically CCI, in option trading. However, literature shows that prudent use of technical analysis definitely helps trader in enhancing their return and risk in option as well other segment of market and CCI is also one of the best indicators of technical analysis which generate profitable trades.

Rationale of Study

The present study is in area of utilizing technical analysis for option trading. It was found from literature that the evidences are meagre in numbers. Even there are very few published research paper with empirical study which actually describe utility of technical analysis, though technical analysis is most acceptable tool by industry professionals to practice trading in stock market. Moreover, empirical studies on option trading strategy are also very few which actually suggest profit making strategy to option traders, though it always remains point of interest by industry professionals to execute profitable strategies in option trading with minimum investment. Moreover, option trading is perceived as a bit complicated area of study due to complexities in its pricing and very high volatility. Therefore, the present study is an attempt to fulfill this gap by exploring utility of technical analysis for option trading, mainly in creating long position in call and put option, where option trader can generate substantial amount of profit at very calculated risk.

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Research Methodology

The research methodology followed for the research is described as follow:

Aim

The aim of study is to come out with technical trading system based on CCI – An indicator of Technical Analysis for option traders willing to take long positions in NSE NIFTY's call and put options.

Objectives

The objectives of the study are:

- To find out the net return in Option trading as per pre-defined rule of trading system on CCI
- To access the risk zone and possibility of Loss in Option trading as per pre-defined rule of trading system on CCI.
- To evaluate efficiently of trading system based CCI for option trading
- To compare in return Option trading as per pre-defined rule of trading system on CCI with standard return
- To check the consistency in return as per pre-defined rule of trading system on CCI over a period of time

Research Design

The design adopted for the research is descriptive in nature.

Data Collection

The data collected are secondary in nature. The historical data from October 2008 to September 2018 (10 Years) for NSE NIFTY in the money option contracts (on the commencement of month) is collected from official website of National Stock Exchange.

Sampling

The population for study is historical data of NSE NIFTY since its commencement i.e. 2001. However, judgmental sampling is used take data from August 2008 as it was a time when active volume option contract started in the market. Moreover, this time frame includes all the phases of market i.e. bullish trend, bearish trend and a phase of consolidation.

Further, option contract of NSE NIFTY is selected keeping in mind fact that more than two third of the volume in option contracts in India are reported on in this NIFTY index.

As the data collected through official website of national stock exchange, it is found to be reliable and valid for study.

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Hypothesis

Following null hypothesis were framed for the study.

H01 = The pay-off (profit/loss from trades) of proposed trading system in NIFTY Option based in CCI is random in nature.

H02 = There is no significant difference in average profit and average loss occurred in proposed trading system in NIFTY Option based in CCI.

H03 = There is no significant difference between average annual return occurred in proposed trading system in NIFTY Option based in CCI with benchmarked return of 100% per annum.

H04 = There is no significant difference between return from option trades over a period of time

Trading rule

The technical trading system proposed for this study is based on CCI for creating long (Buy) position in call and put NSE NIFTY option contracts. The trading rules are as follow:

Rules of Entry in Option contract

- An option trader long most nearer In-The-Money call option at highest price (In-The-Money call option with nearest strike price) only if all the following condition satisfied
  - Long (Buy) Call option on or between 3rd and 15th day of the respective month with same month expiry date
  - Value of CCI goes above -100.
  - Trade will be done on the day on which the spot NIFTY price will break high of the day on which above "b" indicator generated
- An option trader long most nearer In-The-Money put option at highest price (In-The-Money put option with nearest strike price) only if all the following condition satisfied
  - Long (Buy) Put option on or between 3rd and 15th day of the respective month with same month expiry date
  - Value of CCI goes below 100.
  - Trade will be done on the day on which the spot NIFTY price will break low of the day on which above "b" indicator generated

Rules of Exit from Options contract

A long position in either Call or Put can be squared off (exit) when either of following conditions is satisfied.

- Option price rise by 100% i.e. when gross profit is doubled, or,
- Option price fall by 50% i.e. when gross loss is half, or,

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- On closing price of second last day of expiry date of the respective option contract (if a or b is not achieved)

Assumptions

The assumption of research which are based on general observation during trading in NIFTY Future as follow:

- Lot size for Option contract is 50
- The amount of brokerage is Rs. 100 with all the taxes.
- The slippage cost of trading is 10%

Tools and Techniques used

The mathematical and statistical tools used during study are as follow:

- Commodity Channel Index
- Strike Rate i.e. how much out of total trading signals is profitable.
- Average Gain to Average Loss Ratio (AGAL) i.e. how many times successful trade generate profit against a loss making trade.
- Chi-Square Test to check significance of strike rate.
- Two sample independent T test to check the significance of AGAL.
- One Sample T Test for comparing return with benchmarked return.
- Analysis of Variance for checking consistency in return over a period of time.

MS Excel 2010 and IBM SPSS 20 were used for above calculation. For Willim %R levels, technical charts from http://economictimes.indiatimes.com/markets/technical-charts are used.

Data Analysis

To develop a systematic approach for trading in option contracts of NIFTY based on CCI historical data from 1st October 2008 to 30th September 2018 were collected as per trading rules laid down. As per stipulated rule of entry and exit for creating long position of option contracts, there are 68 trades which are generated in last 10-year time frame. Out of these 68 trades, 42 positions for long put options and 26 positions for long call options.

For every trade net profit / loss are calculated after considering lot size of 50, brokerage & taxes worth Rs. 100 and slippage cost @ 10%. Moreover, time frame for each of the trades is also calculated by taking difference between entry date and exit date into the respective trade.

The Table - I (Enclosed as Annexure - A) presents the payoff of trades with consisting of entry date, entry price, types of options, strike price of nearest ITM options, gross and net profit/loss and time frame for every trade.

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The efficiency of the present option trading system is measured based on its Return, Strike Rate, Average Gain to Average Loss Ratio, Average time frame of Trade, maximum loss zone and annual return of the system. The following Table – II presents the highlights of the same.

Table: II
Option Trading Summary based on Technical Trading System of CCI

Sr. No.	Particular	Value/Nos.
1	Time Frame	10 Years
2	Total Trades	68
3	Average Frequency of Trade	Every 1.76 Month (53 Days)
4	Range of Annual Trade	5 (Min) - 11 (Max.)
5	Overall Profit (Net)	Rs. 1,45,362/-
6	Average Annual Profit (Net)	Rs. 14,536/-
7	Average Investment	Rs. 6,261/-
8	Range of Investment	Rs. 2740/- (Min) – Rs. 15,490 (Max)
9	Average RoI	232.16 % Per Annum
10	Total Profitable Trades	43 out of total 68 trades
11	Average Profit/Gain	Rs. 5150/-
12	Average Profit/Gain in Percentage	81.83%
13	Range of Profit	Rs. 305/- (Min) – Rs. 13841/- (Max)
14	Total Loss making Trades	25 out of total 68 trades
15	Average Loss	Rs. 3043 / -
16	Average Loss in Percentage	51.87%
17	Range of Loss	Rs. 430/- (Min) – Rs. 5738 (Max)
18	Strike Rate	63.25%
19	Average Gain to Average Loss Ratio (AGAL)	1.69 times
20	Consecutive number of profitable trades	4 times
21	Consecutive number of loss making trades	2 times
22	Average Holding Period	9 Days
23	Range of Holding period	1 Day (Min) – 23 Days (Max)

From Table – II, it was found that the system is generating profit of Rs. 1,45,362/- against average investment of Rs. 6261/-. Thus, it can give around 232.16% of profit annually. The strike rate is 63.25% with Average Gain to Average Loss ratio of 1.69 times. It shows that out of every 100 trades around 63 trades are profitable and the profitable trade gives 1.69 times more return than every loss making trade. It was also found that this system will generate trades at an interval of 53 days (1.7 Months) and 4 to 11 trades in a year and there is possibility of 4 consecutive profitable are profitable as well 2 loss making trade.

To check the statistical significance of efficiency of trading system, given hypothesis are tested. To check significance Strike Rate chi-square test is performed to check association between actual profitable trades (i.e. 42/68) & loss making trades (i.e. 25/68) against its random behavior (i.e. 34/68). The following Table – III shows result of chi-square Test.

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Table: III - Chi-Square Test

	Observed	Expected	Test Statistics	Df	p value
Profitable Trades	43 (63.25%)	34 (50%)	4.765	1	0.029
Loss Making Trades	25 (36.75%)	34 (50%)			
Total	80 (100%)	68 (100%)			

p value is significant at 0.05 (5%)

From above table, it is observed that the test result is found statistically significant with p value of 0.029 for chi-square test statistics 4.765 for degree of freedom 1. Thus, the null hypothesis (H01) i.e. "The pay-off (profit/loss from trades) of proposed trading system in NIFTY Option based in CCI is random in nature" is rejected. Therefore, it is inferred that the strike rate of trading system is not random and hence it is significant.

Further, to check statistical significance of Average Gain to Average Loss Ratio, two sample independent T test is carried out to compare Average Net Profit/Gain (After deducting brokerage, tax and slippage cost from Gross Profit) i.e. Rs. 5150/- and Average Net Loss (After adding brokerage, tax and slippage cost in Gross Loss) i.e. Rs. 3043/-. The following Table – IV shows result of two sample independent T Test.

Table: IV – Two Sample Independent T Test

	Mean Value	Test Statistics	Df	p value
Profit per Trade	5150 (81.83%)	29.660	66	<0.001
Loss per Trade	3043 (51.87%)			

p value is significant at 0.05 (5%)

From above table it is found that the test result is found statistically significant with p value of <0.001 for Independent T test statistics 29.660 for degree of freedom 66. Thus, the null hypothesis (H02) i.e. "There is no significant difference in average profit and average loss occurred in proposed trading system in NIFTY Option based in CCI." is rejected. Therefore, it is inferred that the AGAL (Average Gain to Average Loss) Ratio is also statistically significant.

Moreover, the annual return of trading system is also compared with benchmarked return of derivative segment i.e. 100% per annum using one sample T Test. The following Table – V shows result of one sample T Test.

Table: V – One Sample T Test

	Value	Test Statistics	Df	p value
Benchmarked Return per Annum	100%	2.165	68	0.034
System Return per Annum	232.16%			

p value is significant at 0.05 (5%)

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From above table it is found that the test result is found statistically significant with p value of 0.034 for One Sample T test statistics 2.165 for degree of freedom 68. Thus, the null hypothesis (H03) i.e. "There is no significant difference between average annual return occurred in proposed trading system in NIFTY Option based in CCI with benchmarked return of 100% per annum." is rejected. Therefore, it is inferred that the return of proposed trading system is significantly higher than benchmarked return.

Furthermore, to check efficiency of this trading system over a period of time, its return (net profit) is compared within time lags of one year in last 10 years.

Lag I: 3 Equal Lags (Lag of 40 Months each)

Lag II: 5 Equal Lags i.e. an interval of 2 years

Lag III: 10 Equal Lags i.e. an interval of a year.

The consistency in return is compared using One Way Analysis of Variance (ANOVA) for each of the lags. The following Table – VI shows result of one-way ANOVA

Table: VI – One Way Analysis of Variance (ANOVA)

	F Test Statistics	**p value**
Lag – I	0.121	0.948
Lag – II	0.559	0.731
Lag – III	0.594	0.797

p value is significant at 0.05 (5%)

From above table, it is observed that the tests results are found statistically insignificant with p value of more than 0.05 for F test statistics in case of all three time lags. Thus, the null hypothesis (H04) i.e. "There is no significant difference between return from option trades over a period of time" is failed to reject. Therefore, it is inferred that the return of proposed trading system is consistent over a period of time.

Findings

From the analysis and interpretation of historical data of 10 years, it was found out that the technical trading system based CCI produce consistent acceptable results in doing trading in option by creating long positions in both the trend i.e. bullish and bearish. It gives almost 232% return per annum which is significantly higher than benchmarked return of 100% per annum. The Strike rate is found out at 63.25% which is also statistically significant. The AGAL ratio is 1.69 times in which average profit is significantly higher than average loss per trade. Thus, trading system is found to be very efficient and hence recommended for practice.

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It should be noted that the maximum risk is around 58.65% of initial investment and there is possibility of 2 consecutive loss making trade. So, an option trader must have provision to bear loss of 117.30% at a time in worst situation. Thus, if investment in trade is Rs. 100, he/she must have Rs. 217.30 in trading account to meet worst situation of loss. Moreover, it is also found that this system is giving call almost 9 – 10 times in year with holding period of 9 days.

Conclusion

Option trading is always remaining challenging task due to complexities involve in its valuations and very high volatility in its price. The technical analysis is a tool of security analysis which assists in price perdition, trend identifications and identifying momentum in price fluctuations. CCI is one of the widely used indicators to determine momentum for volatility in prices. In this research, an attempt is made to explore this highly untouched area of option trading using technical analysis by developing a trading system for Option traders with long positions based CCI. The result of this trading mechanism is found very attractive and can be implemented by traders and investment advisors to trade in option with calculated risk and return with known time frame and with comparatively small amount of investment. This research also opens up a door in doing similar type of study for other stocks and with different technical indicators.

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Annexure – A: Pay Off

Table – I: Pay Off of Trading System based on CCI in NSE Nifty Options Contract (Long Positions)
(From 1st October 2008 to 30th September, 2018: 10 Years)

Trade No.	Date of Entry in Trade	Nifty Spot Price	Option Type	Strike Price	Expiry Date of Option	Entry Price	Lot Size	Inverst-ment	Date of Exit from Trade	Exit Price	Gross Profit / Loss	Slippage Cost @ 10%	Brokera-ge + Tax	Net Profit / Loss	Trade Duration
1	04-Sep-18	11582	Put	11550	27-Sep-18	141.55	50	7077.50	19-Sep-18	283.1	7077.50	707.75	100.00	6269.75	15
2	14-Jun-18	10808	Put	10800	28-Jun-18	108.00	50	5400.00	22-Jun-18	54.00	-2700.00	270.00	100.00	-3070.00	8
3	14-May-18	10801	Put	10800	31-May-18	126.00	50	6300.00	21-May-18	252.00	6300.00	630.00	100.00	5570.00	7
4	03-Apr-18	10245	Call	10250	26-Apr-18	182.35	50	9117.50	23-Apr-18	364.70	9117.50	911.75	100.00	8105.75	20
5	07-Dec-17	10167	Call	10150	28-Dec-17	110.90	50	5545.00	18-Dec-17	221.80	5545.00	554.50	100.00	4890.50	11
6	07-Nov-17	10350	Put	10350	30-Nov-17	103.55	50	5177.50	15-Nov-17	207.10	5177.50	517.75	100.00	4559.75	8
7	03-Oct-17	9860	Call	9850	26-Oct-17	141.60	50	7080.00	13-Oct-17	283.20	7080.00	708.00	100.00	6272.00	10
8	03-Aug-17	10014	Put	10000	31-Aug-17	102.40	50	5120.00	10-Aug-17	204.80	5120.00	512.00	100.00	4508.00	7
9	04-Jul-17	9614	Call	9600	27-Jul-17	121.00	50	6050.00	11-Jul-17	242.00	6050.00	605.00	100.00	5345.00	7
10	07-Jun-17	9664	Put	9650	29-Jun-17	88.00	50	4400.00	19-Jun-17	44.00	-2200.00	220.00	100.00	-2520.00	12
11	06-Apr-17	9261	Put	9250	27-Apr-17	103.00	50	5150.00	24-Apr-17	51.50	-2575.00	257.50	100.00	-2932.50	18
12	12-Dec-16	8170	Put	8200	29-Dec-16	120.75	50	6037.50	23-Dec-16	241.50	6037.50	603.75	100.00	5333.75	11
13	03-Oct-16	8738	Call	8700	27-Oct-16	178.00	50	8900.00	13-Oct-16	89.00	-4450.00	445.00	100.00	-4995.00	10
14	09-Sep-16	8867	Put	8900	29-Sep-16	100.00	50	5000.00	11-Sep-16	200.00	5000.00	500.00	100.00	4400.00	2
15	10-Aug-16	8575	Put	8600	25-Aug-16	97.40	50	4870.00	16-Aug-16	48.70	-2435.00	243.50	100.00	-2778.50	6
16	08-Jun-16	8273	Put	8300	30-Jun-16	127.00	50	6350.00	24-Jun-16	254.00	6350.00	635.00	100.00	5615.00	16
17	05-Apr-16	7603	Put	7600	28-Apr-16	123.00	50	6150.00	11-Apr-16	61.50	-3075.00	307.50	100.00	-3482.50	6
18	04-Jan-16	7791	Put	7800	28-Jan-16	135.00	50	6750.00	11-Jan-16	270.00	6750.00	675.00	100.00	5975.00	7
19	15-Dec-15	7701	Call	7700	31-Dec-15	114.00	50	5700.00	29-Dec-15	228.00	5700.00	570.00	100.00	5030.00	14
20	13-Oct-15	8132	Put	8150	29-Oct-15	136.00	50	6800.00	16-Oct-15	68.00	-3400.00	340.00	100.00	-3840.00	3

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Trade No.	Date of Entry in Trade	Nifty Spot Price	Option Type	Strike Price	Expiry Date of Option	Entry Price	Lot Size	Inverst-ment	Date of Exit from Trade	Exit Price	Gross Profit / Loss	Slippage Cost @ 10%	Brokera-ge + Tax	Net Profit / Loss	Trade Duration
21	08-Sep-15	7688	Call	7650	24-Sep-15	209.00	50	10450.00	23-Sep-15	203.00	-300.00	30.00	100.00	-430.00	15
22	15-Jun-15	8014	Call	8000	25-Jun-15	128.40	50	6420.00	22-Jun-15	256.80	6420.00	642.00	100.00	5678.00	7
23	08-May-15	8191	Call	8200	28-May-15	162.40	50	8120.00	27-May-15	131.00	-1570.00	157.00	100.00	-1827.00	19
24	15-Apr-15	8750	Put	8750	30-Apr-15	93.70	50	4685.00	20-Apr-15	187.40	4685.00	468.50	100.00	4116.50	5
25	05-Mar-15	8938	Put	8950	26-Mar-15	151.95	50	7597.50	13-Mar-15	303.90	7597.50	759.75	100.00	6737.75	8
26	08-Jan-15	8235	Call	8200	29-Jan-15	155.80	50	7790.00	15-Jan-15	311.60	7790.00	779.00	100.00	6911.00	7
27	08-Dec-14	8538	Put	8550	24-Dec-14	113.75	50	5687.50	12-Dec-14	227.50	5687.50	568.75	100.00	5018.75	4
28	10-Sep-14	8094	Put	8100	25-Sep-14	76.90	50	3845.00	16-Sep-14	153.80	3845.00	384.50	100.00	3360.50	6
29	15-Jul-14	7527	Call	7500	31-Jul-14	110.45	50	5522.50	24-Jul-14	220.90	5522.50	552.25	100.00	4870.25	9
30	08-Jul-14	7623	Put	7650	31-Jul-14	162.90	50	8145.00	11-Jul-14	81.45	-4072.50	407.25	100.00	-4579.75	3
31	13-Jun-14	7542	Put	7550	26-Jun-14	86.75	50	4337.50	17-Jun-14	43.38	-2168.75	216.88	100.00	-2485.63	4
32	09-May-14	6858	Call	6850	29-May-14	309.80	50	15490.00	26-May-14	619.60	15490.00	1549.00	100.00	13841.00	17
33	07-Feb-14	6063	Call	6050	26-Feb-14	125.00	50	6250.00	13-Feb-14	62.50	-3125.00	312.50	100.00	-3537.50	6
34	12-Dec-13	6237	Put	6250	26-Dec-13	74.35	50	3717.50	20-Dec-13	37.15	-1860.00	186.00	100.00	-2146.00	8
35	05-Nov-13	6253	Put	6250	28-Nov-13	98.75	50	4937.50	12-Nov-13	197.50	4937.50	493.75	100.00	4343.75	7
36	12-Aug-13	5612	Call	5600	29-Aug-12	123.00	50	6150.00	16-Aug-13	61.50	-3075.00	307.50	100.00	-3482.50	4
37	11-Apr-13	5594	Call	5600	25-Apr-13	79.40	50	3970.00	18-Apr-13	158.80	3970.00	397.00	100.00	3473.00	7
38	05-Mar-13	5784	Call	5700	28-Mar-13	137.00	50	6850.00	8-Mar-13	274.00	6850.00	685.00	100.00	6065.00	3
39	09-Jan-13	5972	Put	6000	31-Jan-13	78.00	50	3900.00	14-Jan-13	39.00	-1950.00	195.00	100.00	-2245.00	5
40	11-Dec-12	5899	Put	5900	27-Dec-12	64.00	50	3200.00	18-Dec-12	32.00	-1600.00	160.00	100.00	-1860.00	7
41	09-Nov-12	5686	Put	5700	29-Nov-12	63.30	50	3165.00	16-Nov-12	126.60	3165.00	316.50	100.00	2748.50	7
42	05-Oct-12	5746	Put	5800	25-Oct-12	120.65	50	6032.50	23-Oct-12	102.50	-907.50	90.75	100.00	-1098.25	18
43	07-Sep-12	5342	Call	5300	27-Sep-12	103.00	50	5150.00	14-Sep-12	206.00	5150.00	515.00	100.00	4535.00	7
44	11-Jul-12	5306	Put	5300	26-Jul-12	62.95	50	3147.50	18-Jul-12	125.90	3147.50	314.75	100.00	2732.75	7
45	05-Jun-12	4863	Call	4800	28-Jun-12	160.20	50	8010.00	11-Jun-12	320.40	8010.00	801.00	100.00	7109.00	6

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Trade No.	Date of Entry in Trade	Nifty Spot Price	Option Type	Strike Price	Expiry Date of Option	Entry Price	Lot Size	Inverst-ment	Date of Exit from Trade	Exit Price	Gross Profit / Loss	Slippage Cost @ 10%	Brokera-ge + Tax	Net Profit / Loss	Trade Duration
46	09-Mar-12	5333	Call	5300	29-Mar-12	165.00	50	8250.00	19-Mar-12	330.00	8250.00	825.00	100.00	7325.00	10
47	08-Nov-11	5289	Put	5300	24-Nov-12	114.00	50	5700.00	15-Nov-11	228.00	5700.00	570.00	100.00	5030.00	7
48	07-Oct-11	4888	Call	4900	25-Oct-12	150.80	50	7540.00	24-Oct-11	202.00	2560.00	256.00	100.00	2204.00	17
49	08-Jul-11	5660	Put	5700	28-Jul-12	108.00	50	5400.00	25-Jul-11	54.00	-2700.00	270.00	100.00	-3070.00	17
50	03-Jun-11	5516	Put	5500	30-Jun-12	94.40	50	4720.00	20-Jun-11	188.80	4720.00	472.00	100.00	4148.00	17
51	08-Apr-11	5842	Put	5900	28-Apr-12	130.70	50	6535.00	13-Apr-11	65.35	-3267.50	326.75	100.00	-3694.25	5
52	14-Feb-11	5456	Call	5400	24-Feb-11	103.60	50	5180.00	21-Feb-11	51.80	-2590.00	259.00	100.00	-2949.00	7
53	05-Jan-11	6080	Put	6100	27-Jan-11	105.65	50	5282.50	7-Jan-11	211.30	5282.50	528.25	100.00	4654.25	2
54	13-Dec-10	5944	Call	5900	27-Dec-10	126.35	50	6317.50	29-Dec-10	164.45	1905.00	190.50	100.00	1614.50	16
55	12-Nov-10	6194	Put	6200	25-Nov-10	163.00	50	8150.00	19-Nov-10	326.00	8150.00	815.00	100.00	7235.00	7
56	08-Oct-10	6103	Put	6100	28-Oct-10	109.40	50	5470.00	13-Oct-10	54.70	-2735.00	273.50	100.00	-3108.50	5
57	10-Aug-10	5461	Put	5500	26-Aug-10	54.80	50	2740.00	18-Aug-10	27.40	-1370.00	137.00	100.00	-1607.00	8
58	12-Apr-10	5340	Put	5400	29-Apr-10	126.50	50	6325.00	28-Apr-10	171.00	2225.00	222.50	100.00	1902.50	16
59	08-Feb-10	4760	Call	4700	25-Feb-10	170.00	50	8500.00	24-Feb-10	179.00	450.00	45.00	100.00	305.00	16
60	11-Jan-10	5249	Put	5300	28-Jan-10	119.00	50	5950.00	22-Jan-10	238.00	5950.00	595.00	100.00	5255.00	11
61	06-Nov-09	4796	Call	4800	26-Nov-09	134.00	50	6700.00	16-Nov-09	268.00	6700.00	670.00	100.00	5930.00	10
62	05-Oct-09	5003	Put	5000	29-Oct-09	140.00	50	7000.00	14-Oct-09	70.00	-3500.00	350.00	100.00	-3950.00	9
63	05-Aug-09	4694	Put	4700	27-Aug-09	205.00	50	10250.00	28-Aug-09	102.50	-5125.00	512.50	100.00	-5737.50	23
64	14-Jul-09	4111	Call	4100	30-Jul-09	130.45	50	6522.50	17-Jul-09	260.90	6522.50	652.25	100.00	5770.25	3
65	12-Jun-09	4583	Put	4600	25-Jun-09	143.00	50	7150.00	17-Jun-09	286.00	7150.00	715.00	100.00	6335.00	5
66	13-Mar-09	2719	Call	2700	26-Mar-09	76.90	50	3845.00	18-Mar-09	143.90	3350.00	335.00	100.00	2915.00	5
67	07-Jan-09	2920	Put	2900	29-Jan-09	166.00	50	8300.00	28-Jan-09	83.00	-4150.00	415.00	100.00	-4665.00	21
68	12-Nov-08	2938	Put	3000	27-Nov-08	167.00	50	8350.00	20-Nov-08	334.00	8350.00	835.00	100.00	7415.00	8

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